

Panel MS-AR(1) on SOE log real GDP per worker, 1970Q1 onward

1 Model

Joint panel Markov-switching AR(1) around a *linear* trend, with three regimes:

$$y_{it} = a_i + g t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}). \quad (3)$$

The trend is linear in calendar time t (year-fraction; g is per year). The intercept a_i is either common across countries or country-specific; the slope g is always common. The cycle z_{it} is a three-regime AR(1). Countries are independent given shared Markov parameters; latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1} . σ and ρ switch with the regime. The median μ is pinned at 0 after ordering. $|\rho| < 0.995$. Likelihood: Hamilton filter per country, summed.

Specifications

1. Common intercept a and common linear slope g (joint MLE).
2. Country intercepts a_i and common g (a_i profiled).

Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters only. A dashed entry is a pinned coefficient.

2 Common intercept and linear trend

SOE sample from 1970Q1: log real GDP per worker. 62 countries, 6924 observations, 1970-Q1–2024-Q3. Fitted countries: 62. Observations: 6924. Log-likelihood: 16652.31. Common intercept $a = -3.8544$ (0.1934). Common slope $g = 0.0124$ (0.0009).

Table 1: Regime AR(1) and transition matrix. Columns are regimes.

	(0)	(1)	(2)
μ	-0.6834 (0.1942)	0.0000 —	1.3119 (0.2051)
σ	0.0109 (0.0002)	0.1086 (0.0050)	0.0242 (0.0004)
ρ	0.9950 (0.0000)	0.9950 (0.0000)	0.9950 (0.0000)
$\Pi_0.$	0.9706 (0.0036)	0.0198 (0.0035)	0.0096 (0.0031)
$\Pi_1.$	0.0714 (0.0192)	0.7053 (0.0386)	0.2233 (0.0376)
$\Pi_2.$	0.0149 (0.0032)	0.0206 (0.0038)	0.9646 (0.0044)

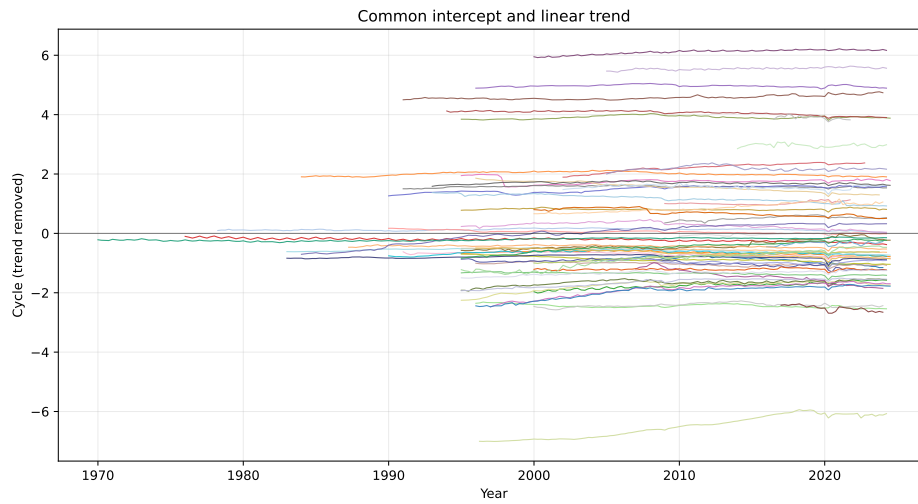


Figure 1: Country cycles after removing the linear trend.

3 Country-specific intercepts, common linear trend

SOE sample from 1970Q1: log real GDP per worker. 62 countries, 6924 observations, 1970-Q1–2024-Q3. Fitted countries: 62. Observations: 6924. Log-likelihood: 17887.41. Country intercepts a_i : mean -3.626, min -10.068, max 2.467. Common slope $g = 0.0127$.

Table 2: Regime AR(1) and transition matrix. Columns are regimes.

	(0)	(1)	(2)
μ	-0.2670	0.0000	0.2165
	—		
σ	0.0224	0.0645	0.0097
ρ	0.9911	0.9166	0.9943
Π_0	0.9441	0.0240	0.0319
Π_1	0.0907	0.8039	0.1054
Π_2	0.0256	0.0129	0.9615

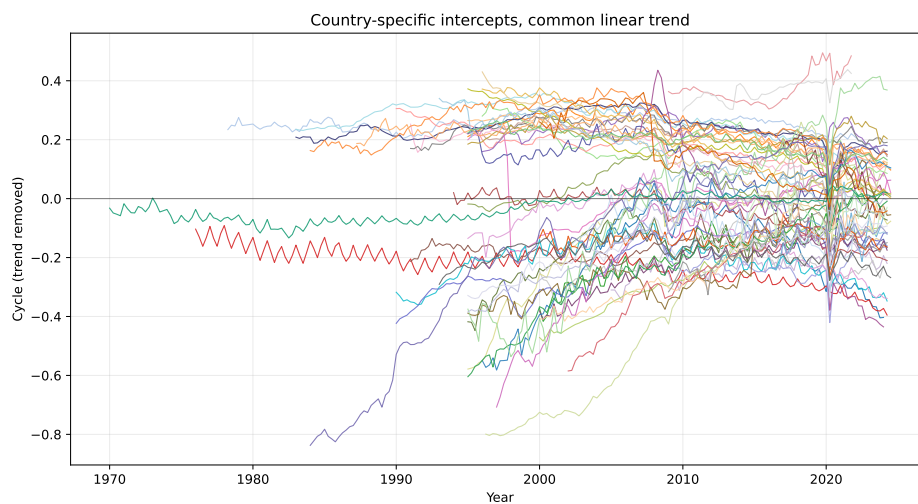


Figure 2: Country cycles after removing the linear trend.